

Patrick Styll

ETH Zürich | pstyll@ethz.ch | patrick-styll.ch | [GitHub](#) | [Google Scholar](#) | [ORCID](#)

EDUCATION

Federal Institute of Technology Zurich (ETH Zürich)

Zurich, Switzerland

M.Sc. Computer Science

Sep 2025 – Aug 2027

- **GPA:** 5.5, with distinction
- **Major:** Machine Intelligence **Minor:** Data Management

Technical University of Vienna (TU Wien)

Vienna, Austria

B.Sc. Software & Information Engineering

Oct 2021 – Jul 2025

B.Hons. in Artificial Intelligence in Medicine

- **GPA:** 1.20, with distinction; top 5%; Dean's List
- Bachelor with Honors — supervised by Prof. Allan Hanbury

RESEARCH EXPERIENCE

Harvard Medical School, Massachusetts General Hospital

Boston, MA, USA

Incoming Master's Thesis Researcher — supervised by Prof. Luca Pinello

Mar 2027 – Aug 2027

- Using single-cell foundation models and sequence-to-function models to understand how genetic variation affects gene expression and cellular states.

ETH Zurich

Basel, Switzerland

Student Researcher — supervised by Prof. Na Cai

Dec 2025 – Aug 2027

- Developing personalized sequence-to-function models for predicting cell-type-specific gene expression from genomic sequence and genotype in single-cell cohorts, outperforming previous SOTA methods.
- Designed probabilistic deep-ensemble models to jointly predict expression across cell-types while estimating both aleatoric and epistemic uncertainty; investigated the biological meaning of the predicted uncertainty.
- Researched the effect of tissue-specific and variant-aware regulatory priors on individualized prediction, and found that they improve the well-known sign-consistency problem of reference-based models.
- Developing probabilistic distillation methods that compress sequence-based predictors into genotype-based linear models while jointly propagating teacher and student uncertainty for uncertainty-aware TWAS.

The University of Tokyo

Tokyo, Japan

Research Assistant — supervised by Prof. Martin C. Frith

Jun 2025 – Aug 2025

- Worked on probabilistic methods for detecting homologous biological sequences using position-dependent models of substitutions, insertions, and deletions.
- Contributed to the development and evaluation of DUMMER (*Dumb Uncomplicated Match ModelER*), a simple method for highly sensitive detection of related sequences from multiple-sequence alignments, resulting in a publication in the Journal of Computational Biology.
- Built a novel profile HMM as well as a novel EM-algorithm, which enabled DUMMER to detect related sequences with even higher sensitivity; presented a poster at RECOMB 2026.

Spanish National Research Council (CSIC)

Madrid, Spain

Research Assistant — supervised by Dr. Leonardo Campillos-Llanos

Feb 2025

- Collected a large corpus of medical texts in 4 languages (English, German, Spanish, French) of both real and AI-generated texts, evaluated several (multi-lingual) methods for text classification and performed XAI methods to investigate cross-lingual terminology transfer.
- Publicly released the text corpus online and led a conference publication.

Seoul National University

Seoul, South Korea

Research Assistant — supervised by Prof. Jiok Cha

Jul 2024 – Sep 2024

- Investigated ICA-based feature extraction from neonatal fMRI brain scans to predict risk of autism/developmental delay. Outperformed prior methods and led a workshop publication.
- Built a new 4D encoder-decoder transformer to predict task-related brain activity from resting-state brain dynamics, and helped in a project on emotion decoding of naturalistic stimuli.

PUBLICATIONS

- [1] Frith, M. C., Boppert, P., **Styll, P.**, and Mirdita, M. “Simple and Thorough Detection of Related Sequences with Position-Varying Probabilities of Substitutions, Insertions, and Deletions.” *Journal of Computational Biology*, 2026.
- [2] **Styll, P.**, Campillos-Llanos, L., Fernández-García, J., and Segura-Bédmar, I. “MedAID-ML: A Multilingual Dataset of Biomedical Texts for Detecting AI-Generated Content.” *CLEF*, 2025.
- [3] Seeliger, M. *, **Styll, P.** *, Staudinger, M., and Hanbury, A. “Human or Not? Light-Weight and Interpretable Detection of AI-Generated Text.” *Working Notes of CLEF*, 2025.
- [4] **Styll, P.** *, Kim, D. *, and Cha, J. “Swin fMRI Transformer Predicts Early Neurodevelopmental Outcomes from Neonatal fMRI.” *International Workshop on Health Intelligence, AAAI*, 2025.
- [5] **Styll, P.**, Kusa, W., and Hanbury, A. “Enhancing Clinical Data Capture: Developing a Natural Language Processing Pipeline for Converting Free Text Admission Notes to Structured EHR Data.” *Workshop on Natural Language for Artificial Intelligence, AI*IA*, 2024.
- [6] **Styll, P.**, Campillos-Llanos, L., Kusa, W., and Hanbury, A. “Cross-Linguistic Disease and Drug Detection in Cardiology Clinical Texts: Methods and Outcomes.” In *Working Notes of CLEF*, 2024.
- [7] Kusa, W., **Styll, P.**, Seeliger, M., Mendoza, Ó. E., and Hanbury, A. “DoSSIER at TREC 2023 Clinical Trials Track.” *The Thirty-Second Text REtrieval Conference (TREC)*, 2023.

AWARDS & SCHOLARSHIPS

Excellence Scholarship	State of Lower Austria (30.000€)	2025 – 2027
UTSIP Scholarship	The University of Tokyo (750€)	2025
Foundation Scholarship	TU Wien (2.500€)	2024
BHons Internship Scholarship	TU Wien / Seoul National University (2.000€)	2024
Excellence Scholarship	TU Wien (1.200€ × 3)	2023 – 2025

WORK EXPERIENCE

Technical University of Vienna <i>Teaching Assistant, Part-Time</i>	Vienna, Austria <i>Oct 2024 – Jun 2025</i>
- Machine Learning (VU) – interviewed students and held an introductory PyTorch workshop for a class of roughly 250 students. Supervised by Prof. Nysret Musliu. - Applied Deep Learning (VU) – graded exercises and held an <u>introductory lecture</u> on Large Language Models for a class of roughly 60 students. Supervised by Dr. Alexander Pacha.	
Prime Force Group <i>IT-Consultant / Full-Stack Developer, Part-Time</i>	Vienna, Austria <i>Jul 2022 – May 2024</i>
- Worked on a large-scale EU-funded project to integrate and standardize Next Generation Sequencing workflows in hospitals for the benefit of cancer patients. - Collaborated with both technicians and doctors to integrate information from cancer gene testing, pharmacogenetics testing and e-medication in proper presentation to medical staff for supporting therapy decision making at bedside.	

VOLUNTEERING

Google Developer Group (TU Wien) <i>Co-Founder and Organizer</i>	Vienna, Austria <i>Sep 2024 – Jun 2025</i>
I co-founded the now largest Google Developer Group in Austria, where I organized bi-weekly journal clubs and speaker series about intelligent systems with the help of my fellow organizers.	
Technical University of Vienna <i>Local Organizing Team</i>	Vienna, Austria <i>Sep 2023</i>
Helped organizing the <u>European Summer School on Information Retrieval</u> in 2023, as part of the Data Science Research Unit under Prof. Allan Hanbury.	